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## “资源智能提取”双学士学位项目培养方案

### Curriculum for Dual Bachelor's Degree program for “Intelligent Resource Extraction”

#### 一、专业介绍

“资源智能提取”双学士学位人才培养项目依托“矿业工程”和“系统科学”两个一级学科，跨工学、理学两个学科门类，由资源与环境工程学院和理学院联合申报实施，面向矿业产业信息化、数智化转型升级发展需求，培养掌握系统的矿物加工、信息与计算科学基础理论，具有家国情怀、国际视野和较强的工程实践能力，具备利用数学思维和方法分析、研究和解决矿产资源利用中的复杂工程问题，能够从事新时代急需的矿物加工大数据采集、分析研究和管理应用的复合型创新人才。

#### I. Professional Introduction

The dual bachelor's degree talent cultivation project of "Intelligent resource extraction" relies on two first level disciplines, "Mining Engineering" and "Systems Science", crossing the categories of engineering and science. It is jointly applied and implemented by the School of Resources and Environment Engineering and the School of Science, aiming to meet the needs of informationization and digital transformation and upgrading of the mining industry. It cultivates composite innovative talents who master the basic theories of mineral processing, information and computational science, have a patriotism, international perspective, and strong engineering practice ability, have the ability to use mathematical thinking and methods to analyze, research, and solve complex engineering problems in mineral resource utilization, and can engage in the collection, analysis, research, and management application of mineral processing big data urgently needed in the new era.

#### 二、培养目标

面向矿冶产业信息化、数智化转型升级发展需求，融合矿物加工工程与信息计算专业优势资源，资源智能提取双学士学位培养德智体美劳全面发展，具有家国情怀、国际视野，系统掌握矿物加工、信息与计算科学基础理论，具备较强的数学思维能力和运用数学方法分析、研究和解决矿产资源利用及相关领域中复杂工程问题的能力，能在矿产资源利用、信息工程、计算科学、经济金融及相关行业领域从事科学技术研究、装备与工程设计、产品与程序开发、企业及生产管理、经济贸易分析等方面工作的高素质复合型创新人才。预期学生毕业5年左右达到以下目标：

1. 具有良好的人文社会科学素养，较强的社会和环境适应能力，良好的表达沟通及协调管理能力、团队意识和合作精神；
2. 社会责任感强，恪守职业道德，具有与工作相关的质量意识、环境保护意识与安全意识；
3. 能综合运用基础理论与专业知识，针对矿产资源利用及相关领域中的复杂工程问题，进行大数据

采集、分析研究、管理应用和程序设计与开发；

4. 具有多学科知识，能熟练运用现代数学与信息工具，胜任矿物加工、钢铁冶金、资源综合利用、计算科学、信息工程、经济金融及相关行业中技术与产品研发、装备与工程设计和生产组织管理等方面的工作。

5. 具有较强的知识学习、信息获取能力和国际视野，能及时了解矿产资源利用、信息与计算科学及相关学科前沿动态与发展趋势，具备终身学习能力。

## **II. Training objectives**

The double major aims to meet the developmental requirement of informatization and digitalization in the mineral and metallurgy engineering. It integrates the advantages of mineral processing engineering and information and computational science, cultivates comprehensive development in morality, intelligence, physical fitness, aesthetics, and labor, possesses patriotism and an international perspective, systematically masters the basic theories of mineral processing, information and computational science, and possesses strong mathematical thinking ability and the application of mathematical methods for analysis. The ability to research and solve complex

engineering problems in the utilization of mineral resources and related fields, and the ability to engage in scientific and technological research, equipment and engineering design, product and program development, enterprise and production management, economic and trade analysis, and other work in the fields of mineral resource utilization, information engineering, computational science, economics and finance, and related industries, are high-quality composite talents. It is expected that graduates will achieve the following goals within five years after graduation:

1. Having good humanities and social science literacy, strong social and environmental adaptability, good communication and coordination management skills, team spirit and cooperation spirit.

2. Strong sense of social responsibility, adherence to professional ethics, and a sense of quality, environmental protection, and safety related to work;

3. Able to comprehensively apply basic theories and professional knowledge, conduct big data collection, analysis and research, management applications, and program design and development for complex engineering problems in mineral resource utilization and related fields;

4. Having interdisciplinary knowledge, proficient in the use of modern mathematics and information tools, competent in work related to mineral processing, steel metallurgy, comprehensive resource utilization, computational science, information engineering, economic and financial industries, as well as technology and product research and development, equipment and engineering design, and production organization management.

5. Having strong knowledge learning, information acquisition ability, and international perspective, able to timely understand the cutting-edge dynamics and development trends of mineral resource utilization, information and computational science, and related disciplines, and have lifelong learning ability.

## **三、专业特色**

“资源智能提取”双学士学位项目，面向矿业产业信息化、数智化转型升级发展需求，突破传统学科专业壁垒，集中矿物加工工程和信息与计算科学两个国家一流本科专业的优势力量，架构融合矿物加工工程和信息与计算科学的学科知识新体系，促进学科深度交叉融合，为学生提供全新的跨学科学习和多样化发展专业平台。

## **III. Professional characteristics**

The dual bachelor's degree program of " Intelligent resource extraction " is aimed at meeting the needs of informationization and digital transformation and upgrading in the mining industry. It breaks through traditional disciplinary barriers, concentrates the advantages of two national first-class undergraduate majors, Mineral Processing Engineering and Information and Computational Science, and constructs a new disciplinary knowledge system that integrates Mineral Processing Engineering and Information and Computational Science, promoting deep interdisciplinary integration and providing students with a new interdisciplinary learning and diversified professional development platform.

#### 四、毕业要求

1. 能够将数学、自然科学、工程基础和专业知用于解决复杂矿物加工工程问题。
2. 能够应用数学、自然科学和工程科学的第一性原理，识别、表达并通过文献研究分析复杂矿物加工工程问题，以获得有效结论。
3. 能够针对复杂矿物加工工程问题开发和设计解决方案，设计满足特定需求的工艺单元、流程或系统，体现创新性，并从全生命周期成本与净零碳要求角度考虑可行性。
4. 能够对复杂矿物加工工程问题进行研究，包括设计实验、分析与解释数据、并通过信息综合得到合理有效的结论。
5. 能够开发、选择与使用恰当的技术、资源、现代工程工具和信息技术工具，对复杂矿物加工工程问题进行预测与模拟，并能够理解其局限性。
6. 在解决复杂矿物加工工程问题时，能够基于工程相关背景知识，分析和评价工程实践对健康、安全、环境、法律以及经济和社会可持续发展的影响，并理解应承担的责任。
7. 具有良好的人文社会科学素养和社会责任感，有工程报国、工程为民的意识，在矿物加工工程实践中遵守工程职业道德、规范和相关法律，履行责任。
8. 能够在多样化、多学科背景下的团队中承担个体、团队成员以及负责人的角色。
9. 能够就复杂矿物加工工程问题与业界同行及社会公众进行有效沟通和交流，包括撰写报告和设计文稿、陈述发言、清晰表达或回应指令，并能考虑到语言、文化和学习背景的差异。
10. 理解并掌握工程项目相关的管理原理与经济决策方法，并能够在多学科环境中应用。
11. 具有自主学习和终身学习的意识和能力，能够理解广泛的技术进步对工程和社会的影响，适应新技术变革，具有批判性思维能力。

#### IV. Requirements

1. Able to apply mathematics, natural sciences, engineering fundamentals, and professional knowledge to solve complex mineral processing engineering problems.
2. Able to apply the first principles of mathematics, natural sciences, and engineering sciences to identify, express, and analyze complex mineral processing engineering problems through literature research, in order to obtain effective conclusions.
3. Able to develop and design solutions for complex mineral processing engineering problems, design process units, processes or systems that meet specific needs, demonstrate innovation, and consider feasibility from the perspective of full lifecycle cost and net zero carbon requirements.
4. Able to conduct research on complex mineral processing engineering problems, including designing experiments, analyzing and interpreting data, and obtaining reasonable and effective conclusions through information synthesis.
5. Able to develop, select, and use appropriate technologies, resources, modern engineering tools, and

information technology tools to predict and simulate complex mineral processing engineering problems, and understand their limitations.

6. When solving complex mineral processing engineering problems, be able to analyze and evaluate the impact of engineering practices on health, safety, environment, law, and sustainable economic and social development based on relevant engineering background knowledge, and understand the responsibilities that should be undertaken.

7. Possess good humanities and social science literacy and a sense of social responsibility, with a sense of engineering for the country and engineering for the people, abide by engineering professional ethics, norms, and relevant laws in mineral processing engineering practice, and fulfill responsibilities.

8. Able to take on the roles of individual, team member, and leader in a diverse and multidisciplinary team.

9. Able to effectively communicate and exchange ideas with industry peers and the general public on complex mineral processing engineering issues, including writing reports and design drafts, presenting speeches, clearly expressing or responding to instructions, and taking into account differences in language, culture, and learning backgrounds.

10. Understand and master the management principles and economic decision-making methods related to engineering projects, and be able to apply them in a multidisciplinary environment.

11. Possess awareness and ability for self-directed and lifelong learning, able to understand the impact of extensive technological advancements on engineering and society, adapt to new technological changes, and possess critical thinking skills.

附：培养目标实现矩阵

|         | 培养目标 1 | 培养目标 2 | 培养目标 3 | 培养目标 4 | 培养目标 5 |
|---------|--------|--------|--------|--------|--------|
| 毕业要求 1  |        |        | √      | √      |        |
| 毕业要求 2  |        |        |        | √      | √      |
| 毕业要求 3  | √      | √      |        |        |        |
| 毕业要求 4  |        |        | √      | √      | √      |
| 毕业要求 5  |        |        | √      | √      |        |
| 毕业要求 6  | √      |        |        |        | √      |
| 毕业要求 7  |        | √      |        |        | √      |
| 毕业要求 8  | √      | √      |        |        |        |
| 毕业要求 9  | √      |        |        | √      |        |
| 毕业要求 10 | √      |        | √      | √      |        |
| 毕业要求 11 |        |        | √      |        | √      |

## 五、专业主干课程

流体力学、分析化学、物理化学、冶金物理化学、固体物料分选学、造块学、矿物加工研究方法、矿物分选综合实验、造块工艺学系统实验、矿物加工工程设计、数学分析、高等代数与解析几何、概率论、常微分方程、数理统计、数学建模、离散数学、运筹学、微分方程数值解、程序设计与算法语言。

## V. Core courses

Fluid Mechanics, Analytical Chemistry, Physical Chemistry, Metallurgical Physical Chemistry, Solid Materials Separation, Agglomeration, Research Techniques of Mineral Processing, Systematic Experiments of Mineral Processing, Systematic Experiments of Agglomeration Technology, Design of Mineral Processing, Mathematical Analysis, Advanced Algebra and Analytic Geometry, Probability Theory, Ordinary Differential Equation, Mathematical Statistics, Mathematical Modeling, Discrete Mathematics, Operations Research, Numerical solution of differential equations, C Programming.

六、基本学制：四年

VI. Recommended length of the program: 4 years

七、授予学位：矿物加工工程工学学士，信息与计算科学理学学士

VII. Degree: Bachelor of Engineering-Mineral Processing Engineering, Bachelor of Science-Information and Computation Science

学生修满所规定的最低毕业学分，符合武汉科技大学授予学士学位规定，授予双学士学位。

八、毕业学分要求：180学分

| 课程类型     |    | 学分要求 | 课程类型     |    | 学分要求 |
|----------|----|------|----------|----|------|
| 1、公共课程平台 |    | 44   | 3、专业课程模块 |    | 57.5 |
| 公共基础课程   |    | 28   | 专业必修课程   |    | 44.5 |
| 通识教育课程   | 必修 | 10   | 专业选修课程   |    | 13.0 |
|          | 选修 | 6    | 4、实践教学模块 |    | 18.5 |
|          |    |      | 专业实践课程   | 必修 | 18.5 |
| 2、学科基础平台 |    | 54   | 5、素质拓展模块 |    | 6    |

\*通识教育选修课6学分包括：理工类、医学类、人文社科类、经济管理类中选择2学分（学生在本专业所属类别外的3个类中选择2个学分）；思想政治及新时代素质教育类选择2学分（“四史”至少选修1个学分）；美育教育类选择2学分（美学和艺术史论类、艺术鉴赏和评论类课程至少选修1个学分）。

## VIII. Credits required for graduation : 160 credits

| Type of courses                  |                  | Academic credits | Type of courses                    |                  | Academic credits |
|----------------------------------|------------------|------------------|------------------------------------|------------------|------------------|
| 1.Common Courses                 |                  | 44               | 3. Specialized Courses             |                  | 57.5             |
| Common Basic Courses             |                  | 28               | Required Courses                   |                  | 44.5             |
| General Education<br><br>Courses | Required Courses | 10               | Elective Courses                   |                  | 13.0             |
|                                  | Elective Courses | 6                | 4.Practicum and Internship Courses |                  | 18.5             |
|                                  |                  |                  | Disciplinary Practical<br>Courses  | Required Courses | 18.5             |
| 2.General Disciplinary Courses   |                  | 54               | 5.Quality Development Courses      |                  | 6                |

## 九、学分比例

### IX. Ratio of Credits

#### 1、必修选修学分比例

The proportion of compulsory elective credits

| 类别 | 学分    | 占总学分比例 |
|----|-------|--------|
| 必修 | 159.5 | 88.6%  |
| 选修 | 20.5  | 11.4%  |

#### 2、实践教学环节学分比例

The Proportion of credits in practice teaching

| 类别     | 学分     | 占总学分比例 |
|--------|--------|--------|
| 实践教学环节 | 实验教学学分 | 21.4   |
|        | 实践教学模块 | 18.5   |
|        | 素质拓展模块 | 6      |
|        |        | 25.5%  |

## 十、辅修

修读本专业辅修课程达到26学分且主修专业达到毕业要求者，颁发辅修专业证书。修读本专业辅修课程达到40学分（含实践教学），并达到辅修学位授予条件的，颁发辅修学位证书。

### X. Minor

The minor certificate will be issued to those who have completed 26 credits of minor courses of their own major and their major has met the graduation requirements. The minor degree certificate will be issued to those who have completed 40 credits of minor courses (including practical teaching) and meet the conditions for awarding minor degree.

## 十一、毕业要求实现矩阵

## XI. Graduation Realization Matrix

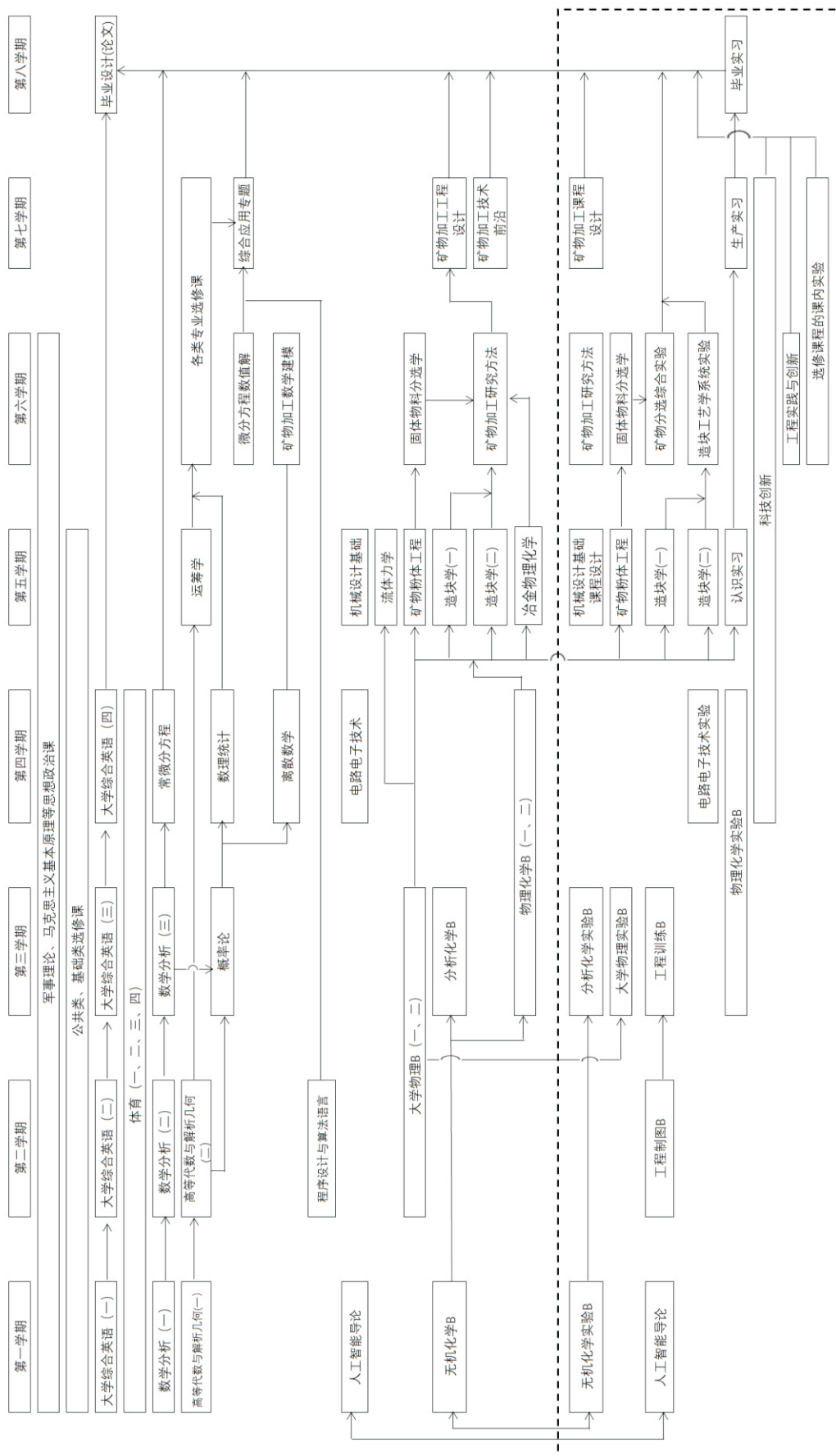
| 课程名称                 | 专业毕业要求 |   |   |   |   |   |   |   |   |    |    |
|----------------------|--------|---|---|---|---|---|---|---|---|----|----|
|                      | 1      | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 思想道德与法治              |        |   |   |   |   | √ | √ |   |   |    |    |
| 中国近现代史纲要             |        |   |   |   |   |   | √ |   |   |    |    |
| 马克思主义基本原理            |        |   |   |   |   |   | √ |   |   |    |    |
| 毛泽东思想和中国特色社会主义理论体系概论 |        |   |   |   |   | √ | √ |   |   |    |    |
| 习近平新时代中国特色社会主义思想概论   |        |   |   |   |   | √ | √ |   |   |    |    |
| 军事理论                 |        |   |   |   |   |   | √ |   |   |    |    |
| 大学生心理健康教育            |        |   |   |   |   |   | √ | √ |   |    | √  |
| 职业生涯规划与就业指导          |        |   |   |   |   |   | √ | √ |   |    | √  |
| 大学综合英语               |        |   |   |   | √ |   |   | √ | √ |    | √  |
| 体育                   |        |   |   |   |   |   | √ | √ |   |    |    |
| 公益劳动                 |        |   |   |   |   |   | √ | √ |   |    |    |
| 形势与政策                |        |   |   |   |   | √ | √ |   | √ |    |    |
| 大学生创新创业基础            |        |   |   |   |   |   | √ |   |   |    | √  |
| 人工智能导论               |        |   | √ |   | √ |   |   | √ |   |    | √  |
| 高等代数与解析几何            | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 数学分析                 | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 概率论                  | √      | √ | √ | √ | √ | √ |   |   |   |    |    |
| 常微分方程                | √      | √ | √ | √ |   |   |   |   |   |    |    |
| 人文社科类公选课             |        |   |   |   |   | √ | √ | √ | √ |    |    |
| 经济管理类公选课             | √      |   |   |   |   | √ |   | √ |   |    |    |
| 医学类公选课               | √      |   |   |   |   |   |   |   |   |    |    |
| 思想政治及新时代素质教育类公选课     |        |   |   |   |   |   | √ |   |   |    |    |
| 美育教育类公选课             |        |   |   |   |   |   | √ |   |   |    |    |
| 工程制图 B               | √      |   | √ | √ | √ |   |   |   |   | √  |    |
| 机械设计基础               | √      |   | √ | √ | √ |   |   |   |   | √  |    |
| 电路电子技术               | √      |   | √ | √ |   |   |   |   |   |    |    |
| 电路电子技术实验             | √      |   | √ | √ |   |   | √ | √ |   |    |    |

| 课程名称       | 专业毕业要求 |   |   |   |   |   |   |   |   |    |    |
|------------|--------|---|---|---|---|---|---|---|---|----|----|
|            | 1      | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 大学物理 B     | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 大学物理实验 B   | √      | √ | √ | √ | √ |   | √ | √ |   |    |    |
| 工程力学 B     | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 无机化学 B     | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 无机化学实验 B   | √      | √ | √ | √ | √ |   | √ | √ |   |    |    |
| 物理化学 B     | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 物理化学实验 B   | √      | √ | √ | √ | √ |   | √ | √ |   |    |    |
| 分析化学 B     | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 分析化学实验 B   | √      | √ | √ | √ | √ |   | √ | √ |   |    |    |
| C 语言程序设计   | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| C 语言程序设计实验 |        | √ | √ | √ | √ |   |   |   |   |    |    |
| 冶金物理化学     | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 流体力学       | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 造块学        |        | √ | √ | √ | √ | √ |   |   | √ |    |    |
| 固体物料分选学    |        | √ | √ | √ | √ | √ |   |   | √ |    |    |
| 资源提取研究方法   |        | √ | √ | √ | √ | √ |   |   | √ | √  | √  |
| 资源提取工程设计   |        | √ | √ | √ | √ | √ |   | √ | √ |    | √  |
| 造块工艺学系统实验  |        | √ | √ | √ | √ |   | √ | √ |   |    | √  |
| 矿物分选综合实验   |        | √ | √ | √ | √ |   | √ | √ |   |    |    |
| 资源提取技术前沿   |        |   |   |   | √ |   |   |   | √ | √  |    |
| 矿物显微结构学    | √      | √ | √ | √ | √ |   |   |   |   |    |    |
| 数理统计       | √      | √ |   | √ |   |   |   |   |   |    |    |
| 矿物加工数学建模   | √      | √ |   | √ | √ |   |   |   |   |    |    |
| 离散数学       | √      | √ |   | √ |   |   |   |   |   |    |    |
| 运筹学        | √      | √ |   | √ |   |   |   |   |   |    |    |
| 微分方程数值解    | √      | √ |   | √ |   |   |   |   |   |    |    |
| 微分方程数值解实验  |        |   |   | √ | √ |   |   |   | √ |    |    |
| 军事训练       |        |   |   |   |   |   | √ | √ |   |    |    |
| 工程训练 B     |        |   | √ |   |   |   | √ | √ |   |    |    |
| 机械设计基础课程设计 | √      |   | √ |   | √ |   | √ | √ |   |    |    |

| 课程名称     | 专业毕业要求 |   |   |   |   |   |   |   |   |    |    |
|----------|--------|---|---|---|---|---|---|---|---|----|----|
|          | 1      | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 |
| 认识实习     | √      | √ | √ | √ |   |   | √ | √ | √ | √  | √  |
| 生产实习     | √      | √ | √ | √ |   | √ | √ | √ | √ | √  | √  |
| 矿物加工课程设计 | √      | √ | √ | √ |   |   | √ | √ |   | √  | √  |
| 毕业实习     | √      | √ | √ | √ |   | √ | √ | √ | √ | √  | √  |
| 毕业设计(论文) | √      | √ | √ | √ | √ | √ | √ | √ | √ | √  | √  |
| 创新创业实践   |        |   |   |   |   |   | √ |   |   | √  |    |
| 第二课堂     |        |   |   |   |   |   | √ | √ |   | √  |    |

## 十二、课程修读进程表

## XII. Course review Process Map



十三、教学环节设置及学分分布表（四年制本科 2025级 “资源智能提取”双学士学位项目）

XIII. Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质 | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                                                    | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双<br>学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------|------------------|---------------------------------------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|----------------------------|----------------|
|                  |          |                  |                                                                                                                     |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                            |                |
| 平台               | 公共基础课程   | 1401048          | 大学综合英语(一)<br>College English (I)                                                                                    | 2      | 32     | 32      | 0   | 0   | 0                | 1      |                            |                |
|                  |          | 1501882          | 体育(一)<br>Physical Education(I)                                                                                      | 1      | 26     | 26      | 0   | 0   | 0                | 1      |                            |                |
|                  |          | 5105017          | 思想道德与法治<br>Ideology and morality and rule of law                                                                    | 3      | 48     | 42      | 0   | 0   | 6                | 1      |                            |                |
|                  |          | 1401072          | 大学综合英语(二)<br>College English (II)                                                                                   | 2      | 24     | 24      | 0   | 0   | 0                | 2      |                            |                |
|                  |          | 1501883          | 体育(二)<br>Physical Education(II)                                                                                     | 1      | 34     | 34      | 0   | 0   | 0                | 2      |                            |                |
|                  |          | 5103001          | 中国近现代史纲要<br>An Outline of Modern and Contemporary History of China                                                  | 3      | 48     | 42      | 0   | 0   | 6                | 2      |                            |                |
|                  |          | 1401049          | 大学综合英语(三)<br>College English (III)                                                                                  | 2      | 32     | 32      | 0   | 0   | 0                | 3      |                            |                |
|                  |          | 1401074          | 大学综合英语(四)<br>College English (IV)                                                                                   | 1      | 16     | 16      | 0   | 0   | 0                | 4      |                            |                |
|                  |          | 1501884          | 体育(三)<br>Physical Education(III)                                                                                    | 1      | 34     | 34      | 0   | 0   | 0                | 3      |                            |                |
|                  |          | 5102001          | 马克思主义基本原理<br>Fundamentals of Marxism                                                                                | 3      | 48     | 44      | 0   | 0   | 4                | 3      |                            |                |
|                  |          | 5106001          | 形势与政策<br>World Affairs and State Policy                                                                             | 2      | 48     | 48      | 0   | 0   | 0                | 1-8    |                            | 分散进行           |
|                  |          | 1501885          | 体育(四)<br>Physical Education(IV)                                                                                     | 1      | 34     | 34      | 0   | 0   | 0                | 4      |                            |                |
|                  |          | 5101005          | 毛泽东思想和中国特色社会主义理论体系概论<br>Theoretical System of Socialism with Chinese Characteristics                                | 3      | 48     | 32      | 0   | 0   | 16               | 4      |                            |                |
|                  |          | 5101006          | 习近平新时代中国特色社会主义思想概论<br>An Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era | 3      | 48     | 32      | 0   | 0   | 16               | 4      |                            |                |
|                  |          | 1501024          | 体质健康测试<br>Physical Health Test                                                                                      | 0      | 16     | 0       | 0   | 0   | 16               | 1-8    |                            | 分散进行           |
|                  | 通识教育课程   | 1306018          | 人工智能导论<br>Introduction to Artificial Intelligence                                                                   | 2      | 32     | 20      | 0   | 12  | 0                | 1      |                            |                |

教学环节设置及学分分布表 (四年制本科 2025级 “资源智能提取” 双学士学位项目)

Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质   | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                           | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|------------|------------------|----------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |            |                  |                                                                            |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 平台               | 公共课程平台课程   | 必修               | 2502006 大学生心理健康教育<br>Mental Health Education                               | 2      | 32     | 24      | 0   | 0   | 8                | 1      |                        |                |
|                  |            |                  | 8001003 大学生创新创业基础<br>Fundamentals of Entrepreneurship                      | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |            |                  | 2503001 职业生涯规划与就业指导<br>Career Planning and Employment Guidance             | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |            |                  | 2504005 军事理论<br>Military Theory                                            | 2      | 36     | 36      | 0   | 0   | 0                | 2      |                        |                |
|                  |            |                  | 5107001 国家安全教育<br>National Security Education                              | 1      | 16     | 16      | 0   | 0   | 0                | 2      |                        |                |
|                  |            |                  | 2501016 大学生劳动教育<br>Labor Education for College Students                    | 1      | 32     | 2       | 0   | 0   | 30               | 1-4    |                        | 分散进行           |
|                  |            | 选修               | 美育教育类 2 学分<br>Art Education2 Academic Credit                               |        |        |         |     |     |                  |        |                        |                |
|                  |            |                  | 医学类 1 学分<br>Medicine1 Academic Credit                                      |        |        |         |     |     |                  |        |                        |                |
|                  |            |                  | 思想政治及新时代素质教育类 2 学分<br>Ideological and Political Education2 Academic Credit |        |        |         |     |     |                  |        |                        |                |
|                  |            |                  | 人文社科类 1 学分<br>Humanity and Social Science1 Academic Credit                 |        |        |         |     |     |                  |        |                        |                |
|                  |            |                  | 经济管理类 1 学分<br>Economic and Management1 Academic Credit                     |        |        |         |     |     |                  |        |                        |                |
|                  |            |                  |                                                                            |        |        |         |     |     |                  |        |                        |                |
|                  | 专业学科基础平台课程 | 必修               | 0702076 高等代数与解析几何(一)<br>Advanced Algebra and Analytic Geometry(I)          | 4      | 64     | 64      | 0   | 0   | 0                | 1      |                        |                |
|                  |            |                  | 0702167 数学分析(一)<br>Mathematical Analysis(I)                                | 4      | 64     | 64      | 0   | 0   | 0                | 1      |                        |                |
|                  |            |                  | 2206665 无机化学B<br>Inorganic Chemistry B                                     | 3      | 48     | 48      | 0   | 0   | 0                | 1      |                        |                |
|                  |            |                  | 2253020 无机化学实验B<br>Experiments in Inorganic Chemistry B                    | 1      | 16     | 0       | 16  | 0   | 0                | 1      |                        |                |
|                  |            |                  | 0305091 工程制图B<br>Engineering Drawing B                                     | 2.5    | 40     | 34      | 0   | 6   | 0                | 2      |                        |                |
|                  |            |                  | 0702170 高等代数与解析几何(二)<br>Advanced Algebra and Analytic Geometry(II)         | 4      | 64     | 64      | 0   | 0   | 0                | 2      |                        |                |

教学环节设置及学分分布表
 (四年制本科 2025级 “资源智能提取” 双学士学位项目)
   
 Offered Course and Distribution of Academic Credits

| 课 程 类 型 |        |          | 课程性质 | 课 程 编 码 | 课 程 名 称                                                             | 学 分 | 合 计 | 课 内 学 时 |     |     | 实践学时 | 学 期 | 是否辅修/双学位课程 | 先 修 课 程/备 注 |
|---------|--------|----------|------|---------|---------------------------------------------------------------------|-----|-----|---------|-----|-----|------|-----|------------|-------------|
|         |        |          |      |         |                                                                     |     |     | 讲 课     | 实 验 | 上 机 |      |     |            |             |
| 平台      | 专业基础平台 | 专业学科基础课程 | 必修   | 0702168 | 数学分析(二)<br>Mathematical Analysis(II)                                | 4   | 64  | 64      | 0   | 0   | 0    | 2   |            |             |
|         |        |          |      | 0703605 | 大学物理B(一)<br>College Physics B(I)                                    | 2.5 | 40  | 40      | 0   | 0   | 0    | 2   |            |             |
|         |        |          |      | 0401039 | 电路电子技术<br>Electric Circuit and Electronic Technologies              | 3   | 48  | 48      | 0   | 0   | 0    | 4   |            |             |
|         |        |          |      | 0450005 | 电路电子技术实验<br>Electric Circuit and Electronic Technologies Experiment | 1   | 16  | 0       | 16  | 0   | 0    | 4   |            |             |
|         |        |          |      | 0702169 | 数学分析(三)<br>Mathematical Analysis(III)                               | 4   | 64  | 64      | 0   | 0   | 0    | 3   |            |             |
|         |        |          |      | 0702171 | 概率论<br>Probability Theory                                           | 3   | 48  | 48      | 0   | 0   | 0    | 3   |            |             |
|         |        |          |      | 0703606 | 大学物理B(二)<br>College Physics B(II)                                   | 2   | 32  | 32      | 0   | 0   | 0    | 3   |            |             |
|         |        |          |      | 0703607 | 大学物理实验B<br>Experiments of College Physics B                         | 1   | 24  | 0       | 24  | 0   | 0    | 3   |            |             |
|         |        |          |      | 2206675 | 物理化学B(一)<br>Physical Chemistry B(I)                                 | 2   | 32  | 32      | 0   | 0   | 0    | 3   |            |             |
|         |        |          |      | 2253054 | 物理化学实验B(一)<br>Experiments in Physical Chemistry B(I)                | 1   | 24  | 0       | 24  | 0   | 0    | 3   |            |             |
|         |        |          |      | 2206681 | 分析化学B<br>Analytical Chemistry B                                     | 2   | 32  | 32      | 0   | 0   | 0    | 3   |            |             |
|         |        |          |      | 2253051 | 分析化学实验B<br>Analytical Chemical Experiment B                         | 1   | 24  | 0       | 24  | 0   | 0    | 3   |            |             |
|         |        |          |      | 0702049 | 常微分方程<br>Ordinary Differential Equation                             | 3   | 48  | 48      | 0   | 0   | 0    | 4   |            |             |
|         |        |          |      | 2206676 | 物理化学B(二)<br>Physical Chemistry B(II)                                | 1.5 | 24  | 24      | 0   | 0   | 0    | 4   |            |             |
|         |        |          |      | 2253026 | 物理化学实验B(二)<br>Experiments in Physical Chemistry B(II)               | 1   | 16  | 0       | 16  | 0   | 0    | 4   |            |             |
|         |        |          |      | 0304017 | 机械设计基础<br>Basics of Mechanical Design                               | 3.5 | 56  | 50      | 6   | 0   | 0    | 5   |            |             |
| 模块      | 专业课程模块 | 专业必修课程   | 必修   | 1306010 | C语言程序设计<br>C Programming                                            | 2.5 | 40  | 40      | 0   | 0   | 0    | 2   |            |             |

教学环节设置及学分分布表 (四年制本科 2025级 “资源智能提取” 双学士学位项目)  
Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                 | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学<br>位课<br>程 | 先 修 课<br>程/备 注 |
|------------------|----------------|------------------|----------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|----------------------------|----------------|
|                  |                |                  |                                                                                  |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                            |                |
| 模块               | 专业<br>课程<br>模块 | 必修               | 1606011<br>C语言程序设计实验<br>Experiment in C Programming                              | 1      | 24     | 0       | 0   | 24  | 0                | 2      |                            |                |
|                  |                |                  | 0702135<br>数理统计<br>Mathematical Statistics                                       | 3      | 48     | 40      | 0   | 8   | 0                | 4      |                            |                |
|                  |                |                  | 0702126<br>离散数学<br>Discrete Mathematics                                          | 4      | 64     | 64      | 0   | 0   | 0                | 4      |                            |                |
|                  |                |                  | 0101194<br>矿物显微结构学<br>Studies of mineral microstructure                          | 2.5    | 40     | 30      | 10  | 0   | 0                | 5      |                            |                |
|                  |                |                  | 0107009<br>流体力学<br>Fluid Mechanics                                               | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                            |                |
|                  |                |                  | 0107065<br>冶金物理化学<br>Metallurgical Physical Chemistry                            | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                            |                |
|                  |                |                  | 0107095<br>矿物粉体工程<br>Mineral powder engineering                                  | 2      | 32     | 28      | 4   | 0   | 0                | 5      |                            |                |
|                  |                |                  | 0107110<br>造块学（一）：烧结理论与工艺<br>Agglomeration(1): Sintering Theory and Techniques   | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                            |                |
|                  |                |                  | 0107111<br>造块学(二)：团矿理论与工艺<br>Agglomeration(2): Pelletizing Theory and Techniques | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                            |                |
|                  |                |                  | 0101027<br>运筹学<br>Operations Research                                            | 2.5    | 40     | 40      | 0   | 0   | 0                | 5      |                            |                |
|                  |                |                  | 0107112<br>资源提取研究方法<br>Research Techniques of Mineral Processing                 | 3      | 48     | 42      | 6   | 0   | 0                | 6      |                            |                |
|                  |                |                  | 0107066<br>矿物分选综合实验<br>Systematic Experiments of Mineral Processing              | 1.5    | 24     | 0       | 24  | 0   | 0                | 6      |                            |                |
|                  |                |                  | 0107067<br>造块工艺学系统实验<br>Systematic Experiments of Agglomeration Technology       | 1.5    | 24     | 0       | 24  | 0   | 0                | 6      |                            |                |
|                  |                |                  | 0107116<br>固体物料分选学<br>Solid Materials Separation                                 | 3.5    | 56     | 50      | 6   | 0   | 0                | 6      |                            |                |
|                  |                |                  | 0702173<br>矿物加工数学建模<br>Mathematical Modeling                                     | 2      | 32     | 24      | 0   | 8   | 0                | 6      |                            |                |
|                  |                |                  | 0702149<br>微分方程数值解<br>Numerical solution of differential equations               | 1.5    | 24     | 24      | 0   | 0   | 0                | 6      |                            |                |

教学环节设置及学分分布表
 (四年制本科 2025级 “资源智能提取”双学士学位项目)

Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程性质   | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称                                                                    | 学<br>分 | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|--------|------------------|-------------------------------------------------------------------------------------|--------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |        |                  |                                                                                     |        |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 模块               | 专业必修课程 | 必修               | 0702150<br>微分方程数值解实验<br>Experiments in Numerical solution of differential equations | 1      | 24     | 0       | 24  | 0   | 0                | 6      |                        |                |
|                  |        |                  | 0107113<br>资源提取工程设计<br>Design of Resource Extraction                                | 3      | 48     | 48      | 0   | 0   | 0                | 7      |                        |                |
|                  |        |                  | 0107122<br>资源提取技术前沿<br>Advanced Resource Extraction Technology                      | 1      | 16     | 16      | 0   | 0   | 0                | 7      |                        |                |
|                  | 专业课程模块 | 选修               | 0702174<br>矿物加工数值分析<br>Numerical Analysis                                           | 3.5    | 56     | 56      | 0   | 0   | 0                | 7      |                        |                |
|                  |        |                  | 0702175<br>矿物加工数值分析实验<br>Numerical Analysis experiment                              | 1      | 16     | 0       | 0   | 16  | 0                | 7      |                        |                |
|                  |        |                  | 0702137<br>基于Python的矿物加工实验与设计<br>Specialized Experiments and Design-based on Python | 2      | 32     | 0       | 32  | 0   | 0                | 7      |                        |                |
|                  |        |                  | 0702176<br>矿业统计计算基础<br>The base of Statistical Computing                            | 1      | 16     | 16      | 0   | 0   | 0                | 6      |                        |                |
|                  |        |                  | 0702177<br>矿业统计计算基础实验<br>Experiments in the base of Statistical Computing           | 1      | 16     | 0       | 0   | 16  | 0                | 6      |                        |                |
|                  |        |                  | 1306012<br>数据库技术及应用<br>Database Technology and Applications                         | 1.5    | 24     | 24      | 0   | 0   | 0                | 4      |                        |                |
|                  |        |                  | 1306013<br>数据库技术及应用实验<br>Experiment of Database Technology and Applications         | 1      | 24     | 0       | 0   | 24  | 0                | 4      |                        |                |
|                  |        |                  | 2206679<br>有机化学B<br>Organic Chemical B                                              | 2.5    | 40     | 40      | 0   | 0   | 0                | 3      |                        |                |
|                  |        |                  | 2253027<br>有机化学实验B<br>Organic Chemical Experiment B                                 | 1      | 24     | 0       | 24  | 0   | 0                | 3      |                        |                |
|                  |        |                  | 0107084<br>中国矿冶科技史<br>History of Mining & Metallurgy in China                       | 2      | 32     | 32      | 0   | 0   | 0                | 4      |                        |                |
|                  |        |                  | 0702025<br>微分几何<br>Differential Geometry                                            | 3      | 48     | 48      | 0   | 0   | 0                | 5      |                        |                |
|                  |        |                  | 0702140<br>统计计算与软件<br>Statistical Computing and Software                            | 2      | 32     | 32      | 0   | 0   | 0                | 5      |                        |                |

## 教学环节设置及学分分布表

(四年制本科 2025级 “资源智能提取”双学士学位项目)

## Offered Course and Distribution of Academic Credits

| 课<br>程<br>类<br>型 | 课程<br>性质       | 课<br>程<br>编<br>码 | 课<br>程<br>名<br>称 | 学<br>分                                                                            | 合<br>计 | 课 内 学 时 |     |     | 实<br>践<br>学<br>时 | 学<br>期 | 是否<br>辅修/<br>双学位<br>课程 | 先 修 课<br>程/备 注 |
|------------------|----------------|------------------|------------------|-----------------------------------------------------------------------------------|--------|---------|-----|-----|------------------|--------|------------------------|----------------|
|                  |                |                  |                  |                                                                                   |        | 讲 课     | 实 验 | 上 机 |                  |        |                        |                |
| 模块               | 专业<br>课程<br>模块 | 专业<br>选修<br>课程   | 0702141          | 统计计算与软件实验<br>Experiments in Statistical Computing<br>and Software                 | 1      | 16      | 0   | 0   | 16               | 0      | 5                      |                |
|                  |                |                  | 0107050          | 矿物化学提取<br>Chemical Extraction of Mineral                                          | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107120          | 矿物加工过程智能化<br>Intelligent measurement and control<br>of mineral processing process | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107102          | 矿物加工专业英语<br>Specialized English                                                   | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0107082          | 智慧环境<br>Wisdom Environment                                                        | 2      | 32      | 32  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0702178          | 计算机网络<br>Computer Network                                                         | 3      | 48      | 48  | 0   | 0                | 0      | 6                      |                |
|                  |                |                  | 0702147          | 回归分析<br>Regression Analysis                                                       | 2      | 32      | 24  | 0   | 8                | 0      | 6                      |                |
|                  |                |                  | 0107052          | 矿物材料<br>Mineral Materials                                                         | 2      | 32      | 32  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 0107056          | 二次资源综合利用<br>Comprehensive Utilization of<br>Secondary Resources                   | 2      | 32      | 32  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 0107087          | 矿物加工工程典型案例<br>Typical Cases Analysis of Mineral<br>Processing Engineering         | 1.5    | 24      | 24  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 0107104          | 3D设计及打印技术<br>3D Design and Printing Technology                                    | 1.5    | 24      | 14  | 0   | 10               | 0      | 7                      |                |
|                  |                |                  | 0107107          | 矿业环境工程<br>Mining Environmental Engineering                                        | 2      | 32      | 32  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 0107108          | 矿物加工分析测试技术<br>Mineral Processing Analysis and<br>Testing Technology               | 2      | 32      | 32  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 0702144          | Java程序设计<br>Java Programming                                                      | 1.5    | 24      | 24  | 0   | 0                | 0      | 7                      |                |
|                  |                |                  | 0702146          | 基于 R 语言的专业实验与设计<br>Specialized Experiments and Design<br>Based on R               | 1.5    | 32      | 0   | 32  | 0                | 0      | 7                      |                |

教学环节设置及学分分布表 (四年制本科 2025级 “资源智能提取” 双学士学位项目)

Offered Course and Distribution of Academic Credits

| 课 程 类 型 |        |        | 课程性质 | 课 程 编 码                                              | 课 程 名 称                                                     | 学 分 | 合 计       | 课 内 学 时 |     |     | 实践学时      | 学 期 | 是否辅修/双学位课程 | 先 修 课 程/备 注 |
|---------|--------|--------|------|------------------------------------------------------|-------------------------------------------------------------|-----|-----------|---------|-----|-----|-----------|-----|------------|-------------|
|         |        |        |      |                                                      |                                                             |     |           | 讲 课     | 实 验 | 上 机 |           |     |            |             |
| 模块      | 实践教学模块 | 专业实践课程 | 必修   | 2504006                                              | 军事训练<br>Military Training                                   | 2   | 112       | 0       | 0   | 0   | 112       | 1   |            |             |
|         |        |        |      | 1701008                                              | 工程训练B<br>Engineering Training B                             | 1.5 | 3<br>(周)  | 0       | 0   | 0   | 3<br>(周)  | 3   |            |             |
|         |        |        |      | 0107030                                              | 认识实习<br>Introductory Practice Experience                    | 1   | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)  | 5   |            |             |
|         |        |        |      | 0304010                                              | 机械设计基础课程设计<br>Course Project in Basics of Mechanical Design | 1   | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)  | 5   |            |             |
|         |        |        |      | 0107031                                              | 生产实习<br>Production Practice                                 | 3   | 3<br>(周)  | 0       | 0   | 0   | 3<br>(周)  | 7   |            |             |
|         |        |        |      | 0107061                                              | 矿物加工课程设计<br>Course Project in Sintering and Pelletizing     | 1   | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)  | 7   |            |             |
|         |        |        |      | 0107096                                              | 毕业设计(论文)<br>Graduation Project (Thesis)                     | 8   | 14<br>(周) | 0       | 0   | 0   | 14<br>(周) | 8   |            |             |
|         |        |        |      | 0107097                                              | 毕业实习<br>Pre-graduation Internship                           | 1   | 2<br>(周)  | 0       | 0   | 0   | 2<br>(周)  | 8   |            |             |
|         | 素质拓展模块 | 创新创业教育 | 必修   | 创新创业实践3学分<br>Innovation Practices 3 Academic Credits |                                                             |     |           |         |     |     |           |     |            |             |
|         |        | 第二课堂   | 必修   | 第二课堂3学分<br>Second Classroom 3 Academic Credits       |                                                             |     |           |         |     |     |           |     |            |             |

\*创新创业教育和第二课堂主要安排在小学期进行

